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STATISTICS

Session 3: Probability & Statistics

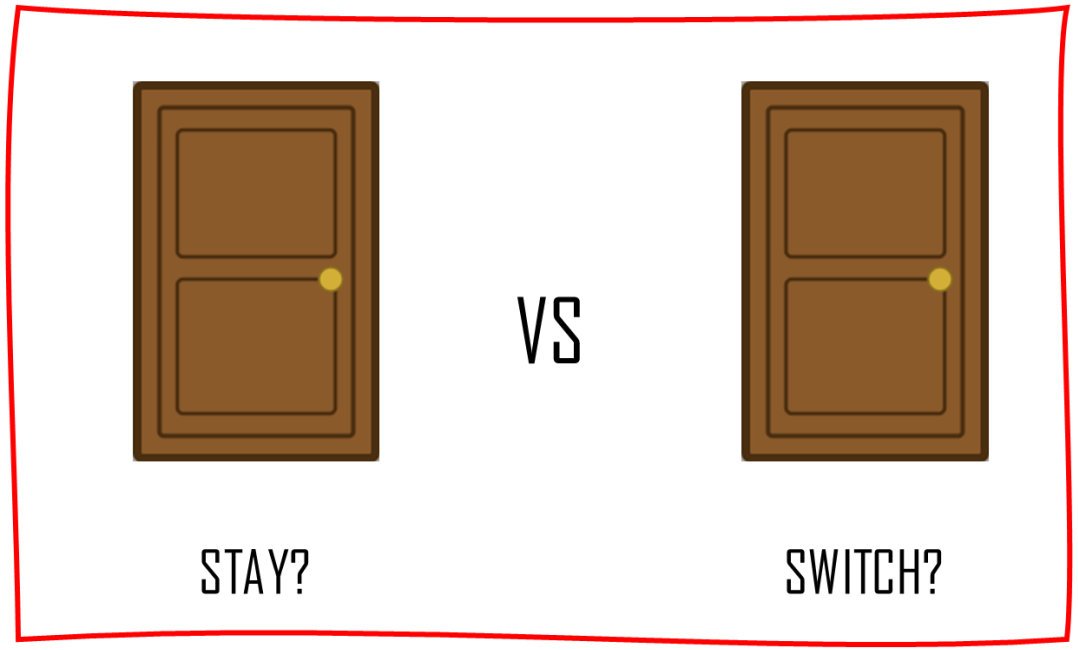
The Monty Hall Problem

Suppose the contestants on a game show are given the choice of three doors: Behind one door is a car; behind the others, goats. After a contestant picks a door, the host, who knows what's behind all the doors, opens one of the unchosen doors, which reveals a goat. He then says to the contestant, "Do you want to switch to the other unopened door, or stay with your current choice?"

Is it to the contestant's advantage to make the switch?

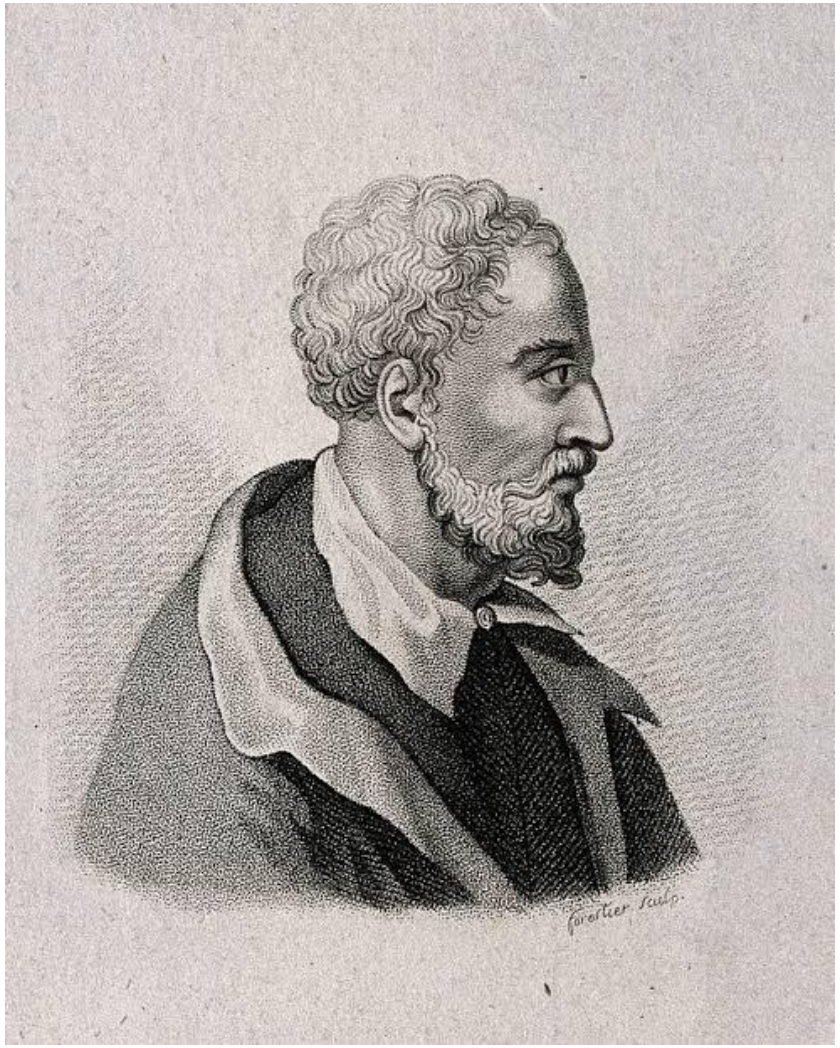


Host reveals one incorrect door



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Cardano & the Sample Space



Mathematics

- Gave the first algebraic solutions for cubic and quartic equations
- Introduced foundation concepts of complex numbers

Physics and Engineering

- Invented the universal joint used to transmit rotary motion
- Described and improved the three-ring concentric gimbal to keep compasses and gyroscopes level
- Contributed to early combination lock designs and hypocycloid "Cardano circles" used in high-speed printing presses

Medicine and Science

- Wrote the first clinical description of typhus fever
- Authored dozens of medical papers

Cardano & the Sample Space

N of heads	Outcomes	
0		$P = \frac{1}{3}$
1		$P = \frac{1}{3}$
2		$P = \frac{1}{3}$

N of heads	Outcomes	
0		$P = \frac{1}{4}$
1		$P = \frac{2}{4}$
		
2		$P = \frac{1}{4}$

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Sample Space Practice Questions

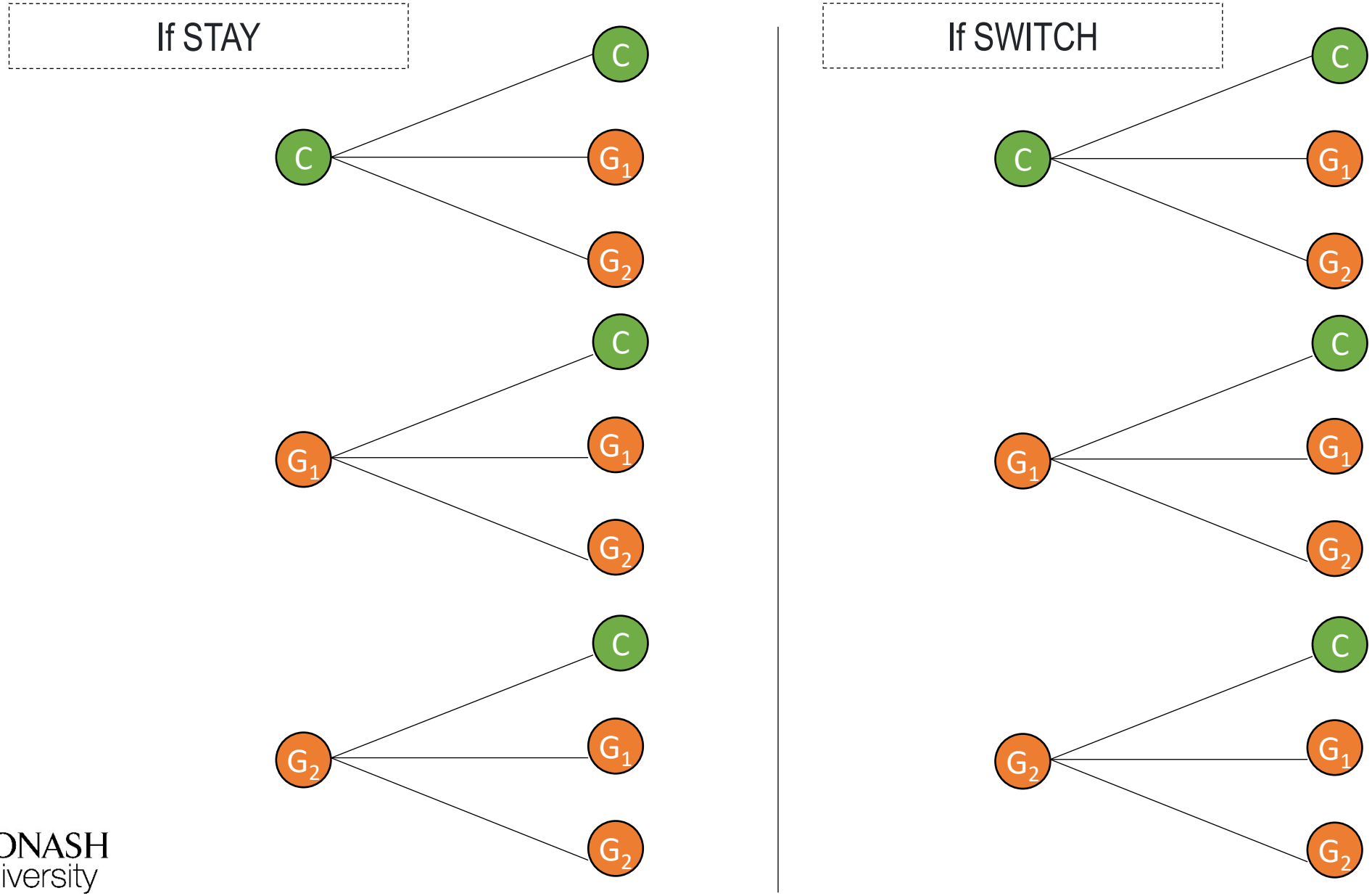
1. Suppose a family has two children, and suppose, for simplicity, that each child is equally likely to be a boy or a girl. What is the probability that both children are girls?

2. Suppose a family has two children, and suppose, for simplicity, that each child is equally likely to be a boy or a girl. What is the probability that at least one of them is a girl?

3. Suppose a family has two children, and that at least one of them is a girl. And suppose, for simplicity, that each child is equally likely to be a boy or a girl. What is the probability that both children are girls?

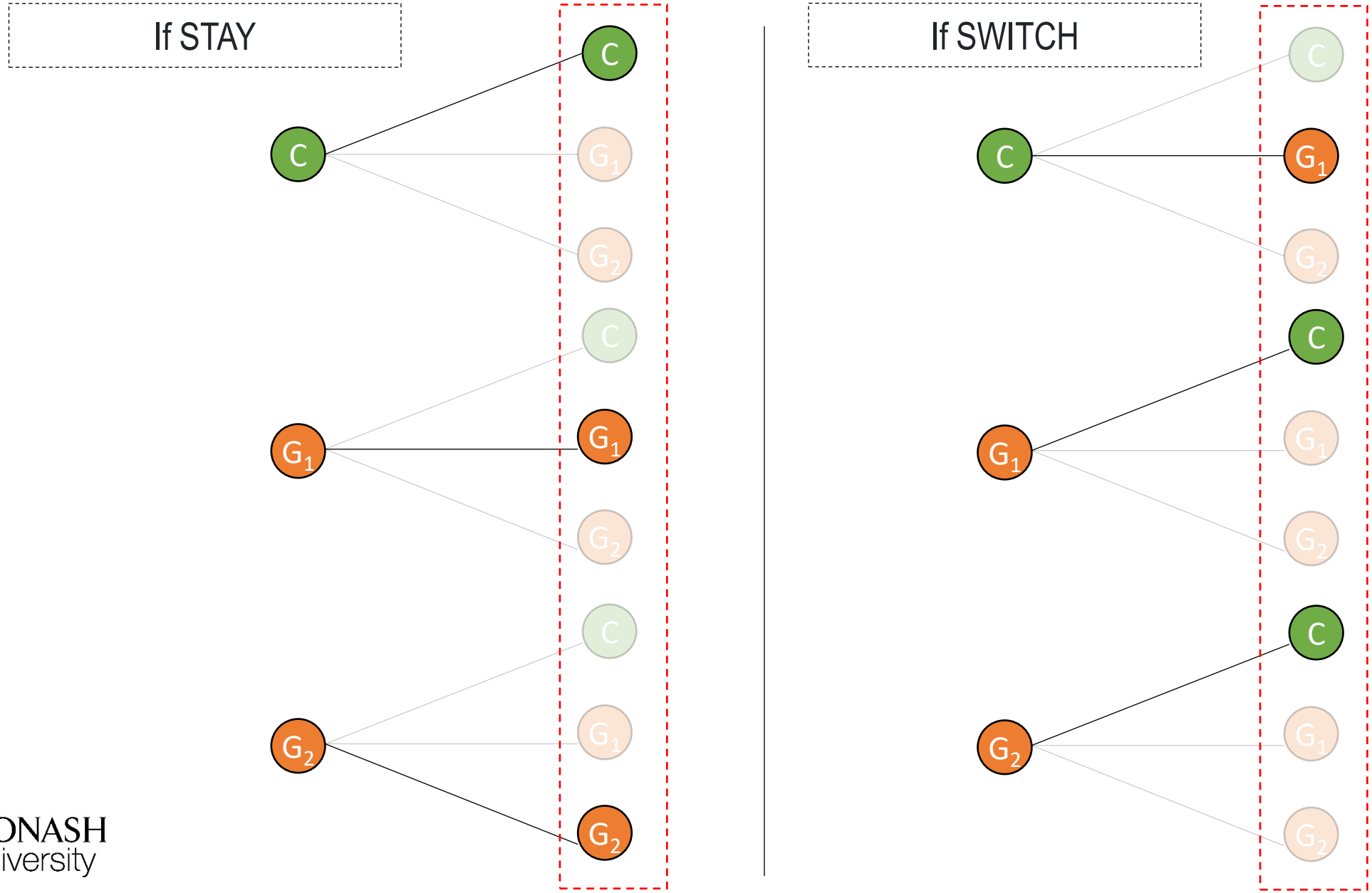
4. Suppose a family has two children, and that the first one is a girl. And suppose, for simplicity, that each child is equally likely to be a boy or a girl. What is the probability that both children are girls?

Solving The Monty Hall Problem



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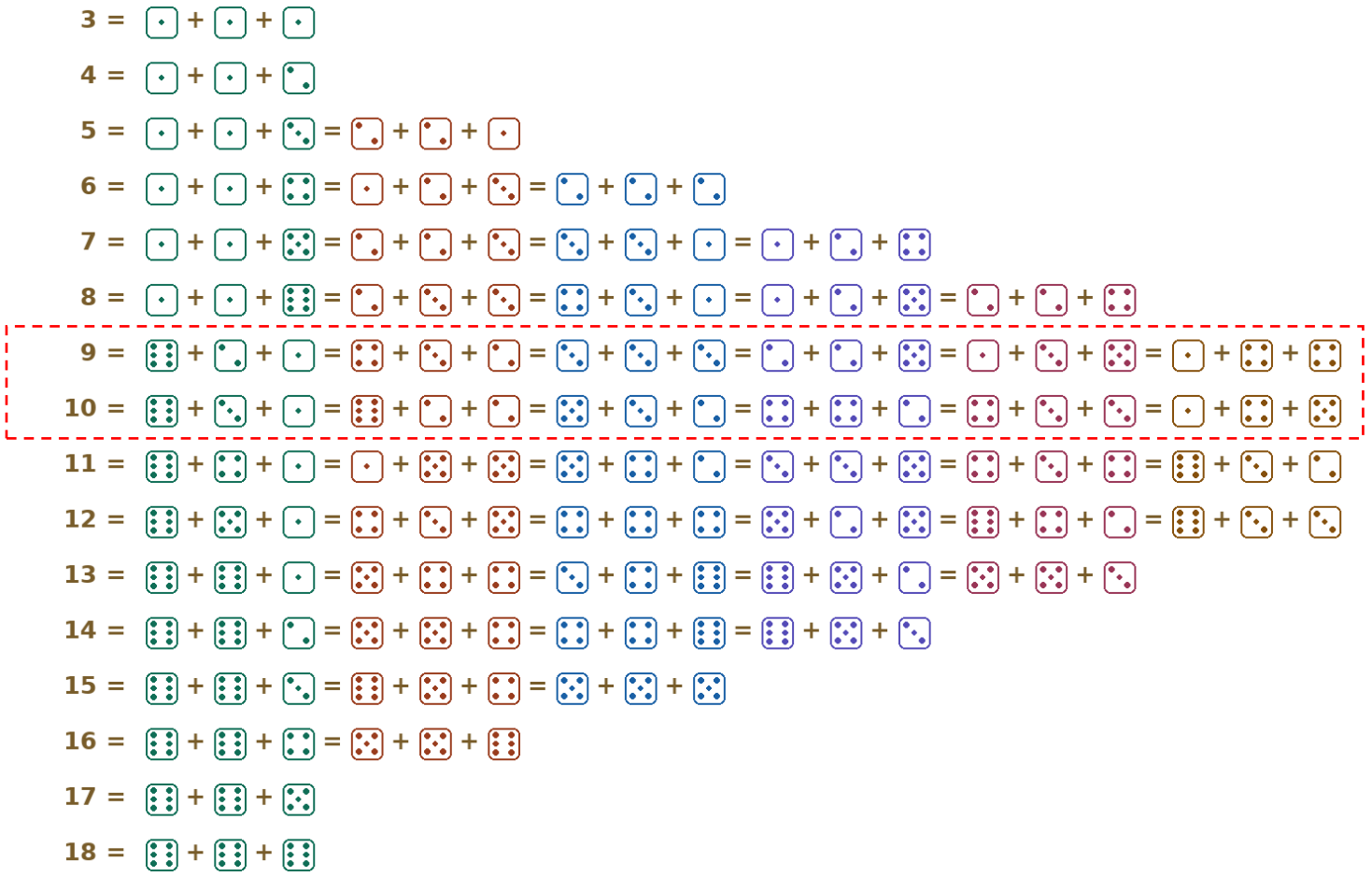
Solving The Monty Hall Problem



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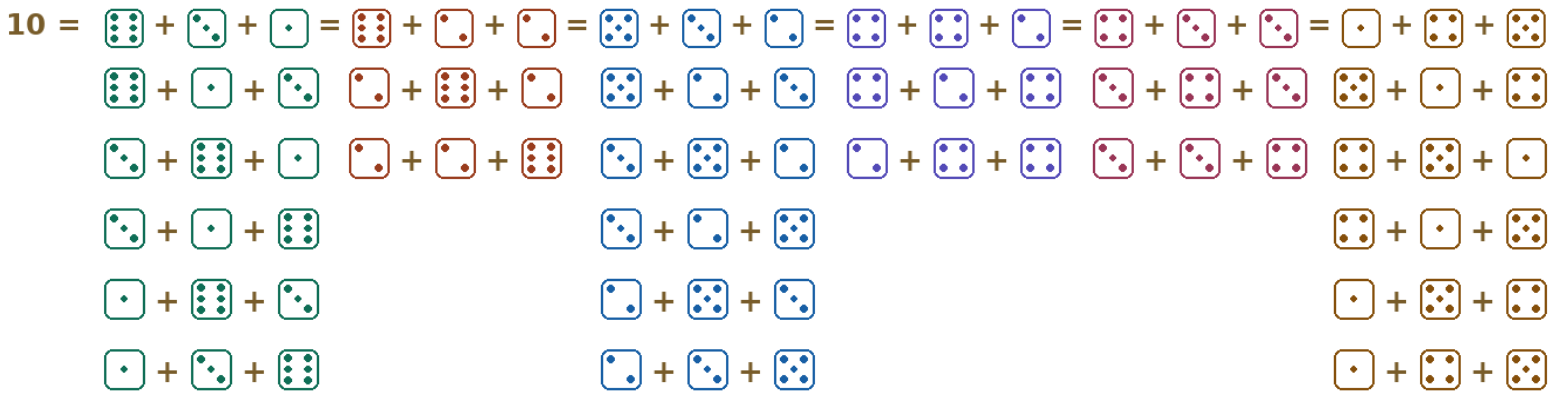
Galileo and Counting

"When you throw 3 dice (and sum the outcomes), why does the number 10 appear more frequently than the number 9?"

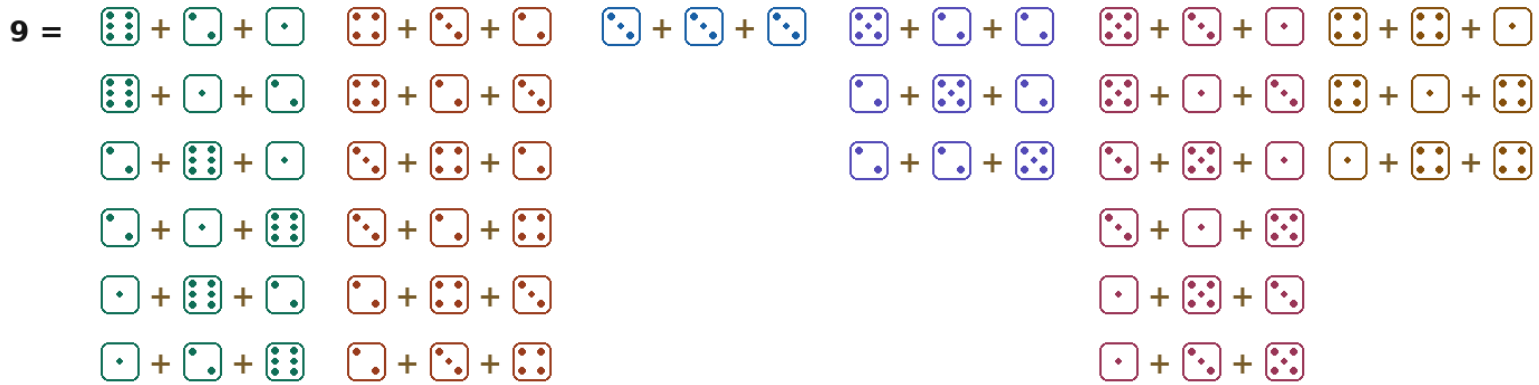


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Galileo and Counting



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Canadian Lottery Example

In the early 2000s the Canadian lottery had 500 cars to award to random subscribers. They used each person's subscriber number as a sampling frame and had approximately 2.4 million numbers. To their embarrassment, the same number cam up twice!

What is the probability of that happening?

Let N = 2,400,000 and n = 500

Assume sampling with replacement

P(at least one repeat) = 1 – P(no repeats)

$$= 1 - \frac{N}{N} \times \frac{N-1}{N} \times \frac{N-2}{N} \times \cdots \frac{N-(n-1)}{N}$$

$$= \prod_{k=0}^{n-1} \left(1 - \frac{k}{N}\right)$$

$$= 0.0507\ (\sim 5\%)$$

Practice Question

How many people do you need in a room for there to be a 50% chance that two share a birthday?

Blaise Pascal



Blaise Pascal (1623 – 1662)

Similar to Galileo, Pascal is remembered for many contributions that are relevant in the modern age:

- The Pascline (mechanical calculator and design for modern calculators)
- Atmospheric Pressure
- The standard unit of Pressure – the pascal (Pa)
- Pascal's principle, which lead to the invention of the hydraulic press
- Pascal's Triangle (used in algebra and combinatorics)
- Towards a working theory of probability

The Problem of Points

Imagine two players are playing a coin-flipping game. We will call them Harry and Tom. Harry scores a point whenever the coin lands Heads (H). Tom scores a point whenever the coin lands Tails (T). The first player to reach 3 points wins the game and takes the prize of $100.

- Now suppose the first three flips are: H, T, H. So the current score is H = 2, and T = 1.
  - Harry needs one more Head to win.
  - Tom needs two more Tails.

- Now imagine that, for some reason, the game is interrupted and cannot continue.
- How should Harry and Tom divide the $100 prize?

The Problem of Points (Wrong Solution)

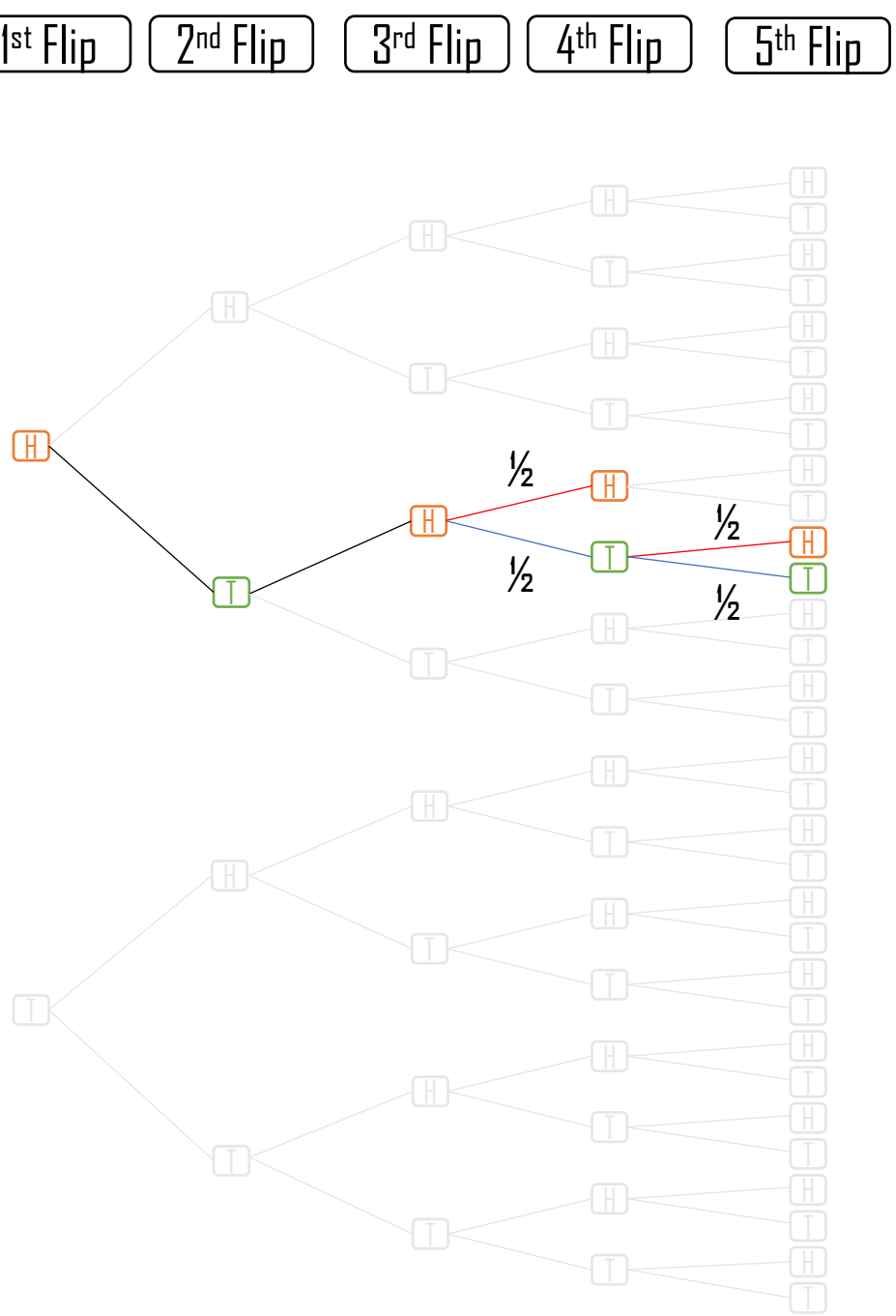
Game:	1	2	3
Outcome:			

- H = Harry's current points = 2
- T = Tom's current points = 1
- n = total points played so far = H + T = 3

Harry's Share = $\frac{H}{H+T} \times 100 = \frac{2}{3} \times 100 = \$66.67$

Tom's Share = $\frac{T}{T+H} \times 100 = \frac{1}{3} \times 100 = \$33.33$

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If we assign probabilitiess to each branch we can determine the chance of each outcome occurring:

Future Flips	Probability	Winner
H	½	Harry
TH	½ x ½ = ¼	Harry
TT	½ x ½ = ¼	Tom

Therefore:

$$P(\text{Harry wins}) = P(H) + P(TH) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$P(\text{Tom wins}) = P(TT) = \frac{1}{4}$$

And the fairest way to split the $100 prize would be:

- Harry = ¾ x 100 = $75
- Tom = ¼ x 100 = $25

Pascal's Triangle

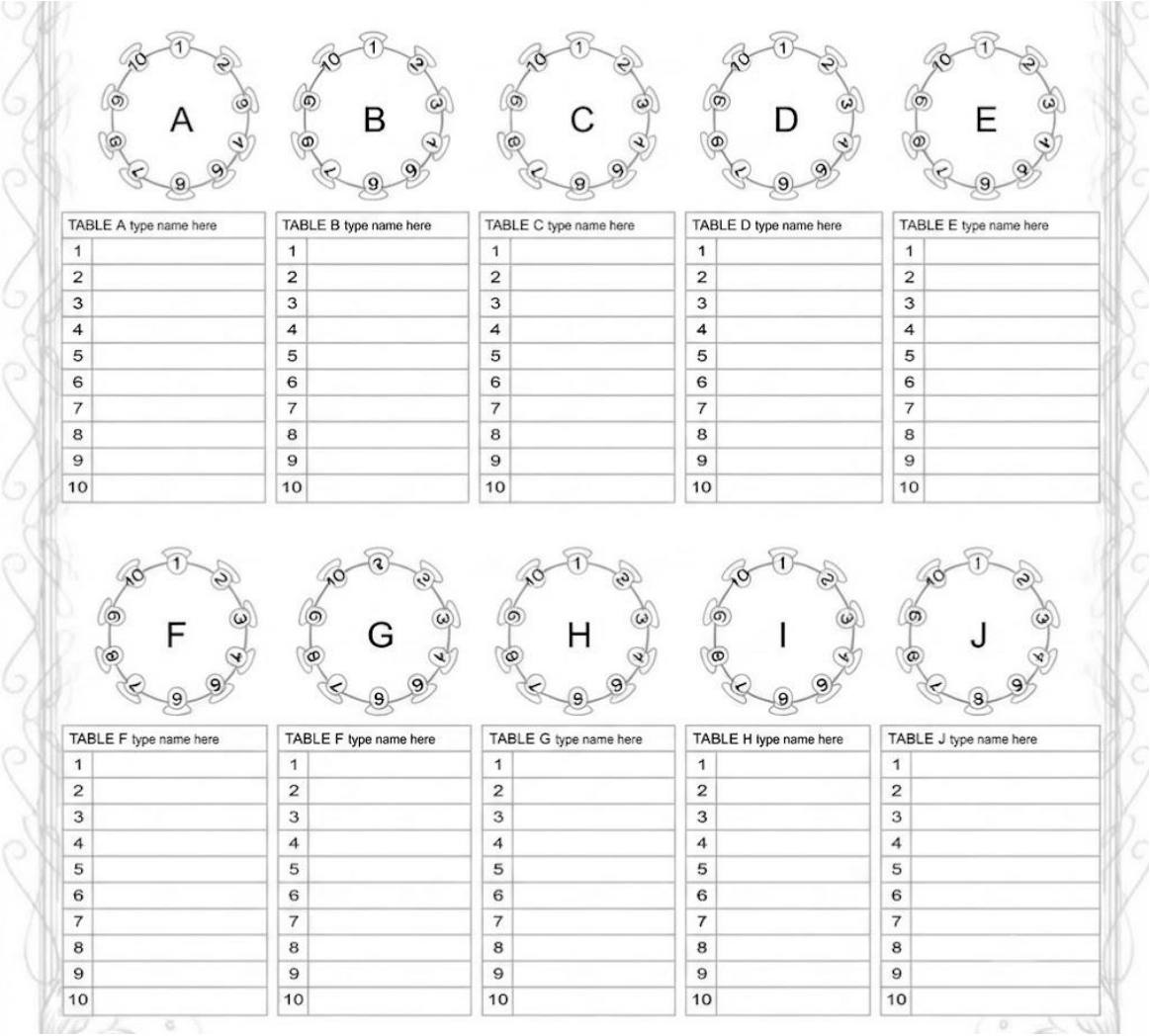


$\binom{4}{0} = 1$ $\binom{4}{1} = 4$ $\binom{4}{2} = 6$ $\binom{4}{3} = 4$ $\binom{4}{4} = 1$

$$1$$
$$1 \quad 1$$
$$1 \quad 2 \quad 1$$
$$1 \quad 3 \quad 3 \quad 1$$
$$1 \quad 4 \quad 6 \quad 4 \quad 1$$
$$1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1$$

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Possible arrangements for a wedding reception



1. Uncle Barry cannot sit near the bar because he becomes argumentative after several drinks
2. Sue's divorced parents must be placed at separate tables, with neither table appearing more important than the other.
3. Aunt Susan should not sit with Cousin Rachel because Susan repeatedly asks when Rachel is going to have children.
4. Uncle Frank and Uncle Peter cannot sit together because they still argue about money borrowed in 2009.
5. Grandma June needs to sit close to the toilets and away from the speakers because she has difficulty walking and dislikes loud music.

...

20. The best man and maid of honour should sit near the newly married couple, but they should not sit beside each other because their partners are both the jealous type.

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Possible arrangements for a wedding reception

$$\binom{100}{10} = \frac{100!}{10!\,(100-10)!}$$

$$= \frac{100!}{10! \times 90!}$$

$$= \frac{100 \times 99 \times 98 \times 97 \times 96 \times 95 \times 94 \times 93 \times 92 \times 91 \times 90!}{10! \times 90!}$$

$$= \frac{62{,}815{,}650{,}955{,}529{,}472{,}000}{3{,}628{,}800}$$

$$= 17{,}310{,}309{,}456{,}440$$

Pascal's Wager – Expected Value

	Heaven exists	Heaven does not exist
Probability	0.5	0.5
Payoff if you lived as a "good" person	$+\infty$ (eternal happiness)	-s (small sacrifice)

$$E = (P_1 \times V_1) + (P_2 + V_2); \quad \text{where P} = \text{Probabilities and V} = \text{Value}$$

$$E(\text{'good' person}) = \underbrace{(0.5)}_{\text{P(heaven exists)}} \times \underbrace{(\infty)}_{\text{Gain}} + \underbrace{(0.5)}_{\text{P(heaven doesn't exists)}} \times \underbrace{(-s)}_{\text{Small loss}}$$

$$= \frac{1}{2}\infty - \frac{1}{2}s$$

$$= \infty$$

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Parking Ticket - Expected Value

	Fine	No Fine
Probability	1/20	19/20
Payoff if you don't pay for parking	-$125	$0
Payoff if you do pay for parking	-$5.50 (certain)	-$5.50 (certain)

Expected Cost of NOT Paying

$$E(\text{payoff}) = \left(\frac{1}{20} \times -125\right) + \left(\frac{19}{20} \times 0\right)$$

$$= -\$6.25$$

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Expected Cost of Paying

$$E(\text{payoff}) = -\$5.50$$

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Australian Lottery - Expected Value



$$\binom{44}{6} = \frac{44!}{6!\,(44-6)!}$$

$$= 7{,}059{,}052$$

$$E(\text{value of ticket}) = \frac{27900000}{7059052} > \text{ticket price} = \$1$$

$$\approx \$3.95$$

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Australian Lottery - Expected Value

Outcome	Frequency	Probability	Share of pot	Payoff
One winner	120 / 170	.706	Full pot	$27.9 million
Two winners (split half)	40 / 170	.235	1/2 pot	$13.95 million
Three winners (split three ways)	10 / 170	.059	1/3 pot	$6.98 million

$$E(\text{winnings}) = \left(\frac{120}{170} \times 27.9\text{M}\right) + \left(\frac{40}{170} \times 13.95\text{M}\right) + \left(\frac{10}{170} \times 6.98\text{M}\right) \approx \$23.4\text{M}$$

$$\frac{E(\text{winnings})}{\text{tickets bought}} \approx \$3.31 \text{ per } \$1 \text{ ticket}$$

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